

Scaling Agile Scrum Software Development

Providing Agility and Quality to Platform Development by Reducing Time to Market

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Abstract—agile software development approach aims at overcoming the limitations of plan-driven software development by allowing requirement changes during all phases of product development and providing agility to organization to respond to changing market needs. Software organizations have successfully implemented agile scrum in distributed software development. However, they also encountered many challenges while implementation which led to considerable amount of effort spend just to manage work. Difficulties were primarily in the areas of communication, culture, different time zones, different level of domain know how across scrum teams, and knowledge management. In this practice paper, we will share practices and systems implemented, challenges encountered along with their countermeasures, and lessons learnt in successfully scaling the Agile Scrum development to 16 globally distributed scrum teams with 100+ team members, successfully delivering 2000+ user stories which required execution of 3000+ product test cases and 1000+ system test cases for verification and validation in a single version of platform release Providing Agility and Quality to Platform Development by Reducing Time to Market.

Keywords—Scaling Scrum; Agile Software Development; Distributed Software Development; Agile Scrum

I. BACKGROUND

The characteristics of Global Software Engineering (GSE) have significant effects on planning, communication, coordination, and control and mitigating the effects is a challenge [1]. Compared to plan-driven software development approaches, agile methods are more flexible when it comes to taking requirements' changes into consideration during all phases of software development [2]. Agile methods emphasize extensive collaboration between customers and developers, and encourage small self-organized collocated teams [3]. Mitigating the Global Software Engineering challenges by themselves is not a straightforward task and combining agile practices with a global or distributed context complicates things even further. The Distance factor in GSE implies a different way of working, organizational standards, organizational cultures and policies, which may decrease a team's cohesion [4].

II. INTRODUCTION

We are working on platform development for one of the Siemens Business unit which can be used as a global product and can also be used as a platform for future building technology applications. Platform development is spread across three continents (Europe, Asia, and North America), over four locations, and 16 Scrum teams. We follow Agile Scrum for Software development. Agile methods, and in particular Scrum [5], have become main-stream in contemporary software development in both small and large organizations [6]. Agile Scrum in general operates broadly around seven basic elements which are defined to meet the expectation of Agile Manifesto. The seven basic elements are (i)Self Organized, Cross Function Collocated Team,(ii)Shorter Iterations of work called "Sprint", (iii)Continuous Monitoring of Progress via "Daily Scrum", (iv)Shippable product every month, (v)Monthly re-evaluation of project goal, (vi) Lean Documentation, (vii) Continuous Integration. The Scrum teams are organized as small teams comprising of experienced developers, testers, analyst and technical writers working as a single team. Other key members of Scrum team are Product Owner, Scrum Master, and management representative also called as stakeholders. Key reason for having collocated team is that business people and developers must work together in close collaboration throughout the projects and should rely on face-to-face communication for resolving issues rather than following the traditional formal methods. Frequent face-to-face communication also improves a feeling of "ONE TEAM" and also builds trust. In this practice paper, we will share our approach for scaling scrum, the challenges we faced and how we overcame and the lessons we learnt during this journey.

III. IMPACT ANALYSIS FOR SCALING SCRUM

Agile Scrum recommends some key practices like maintaining a Product and Sprint Backlog, Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective etc. Since Agile Scrum primarily was designed for small collocated teams, so scaling of Agile Scrum for Globally Distributed Software Engineering (GDSE) will have its impact on agile

scrum practices, and Scrum roles. We identified that the following standard Scrum practices were impacted while scaling Agile for GDSE.

- Product Owner and development team must work together throughout the project, while scaling, although, Product Owner and development team will be operating in different time zones and may not be able to work collaborate face to face.
- The most efficient and effective method of conveying information to and within a development team is face-to-face conversation and here again different scrum teams can work in different time zones.
- Best architectures, requirements, and designs emerge from self-organizing teams. Scaling scrum will have bigger impact over this practice as many scrum teams will be involved and will require careful communication and coordination.
- Scrum Master focuses on removing impediments. While scaling, Scrum Master will invest more time in inter team communication and coordination for dependencies, if any.
- Product Owner closely works with team to meet business objectives. In case of scaling, it can be too overwhelming and challenging for the Product Owner as he/she may be responsible for requirements which will require many Scrum teams to work in different time zones.
- Self-Organized teams define goals and achieve them. Deciding the goal can be challenging and tricky as Scrum team will have to consider dependencies and their resolutions.

IV. THEME FOR SCALING SCRUM

After completing impact analysis of scaling Agile Scrum it was clear that we needed a formidable, clear approach and set of new practices which will help us in overcoming the challenges resulting from Globally Distributed Software Engineering. We moved from traditional Scrum to Hybrid model where we added new roles and new practices which will act as guiding principles to all stakeholders. We grouped our practices broadly under two themes.

- Structure for Scaling Scrum
- Operationalized Scaled Scrum

V. STRUCTURE FOR SCALING SCRUM

Outcome of the impact analysis of scaling Agile Scrum was analyzed and it was clear that since we are in a distributed product development which has software and hardware components, hence we cannot implement pure Agile Scrum. We adopted an in-house developed hybrid model called Product Evolution Process (PEP) along with measures which will help us in establishing strong infrastructure for Communication and Coordination, Maintain High Ownership Team Culture, System for Dependency Identification and Adaptation, Transparency of Planning, Attention to Quality, Common Goal and Proactive and efficient leadership team for quick impediments resolution.

A. Hybrid Model

The PEP agile is an in-house developed hybrid model which consists of Requirement Engineering, System Testing, Field Testing and Release to Market phases defined in

traditional waterfall way, which is sequential; but Software Design, Development and Product Testing, follow the Agile Scrum Practices. Hybrid model starts and ends with waterfall. Agile Scrum is sandwiched in between waterfall phases.

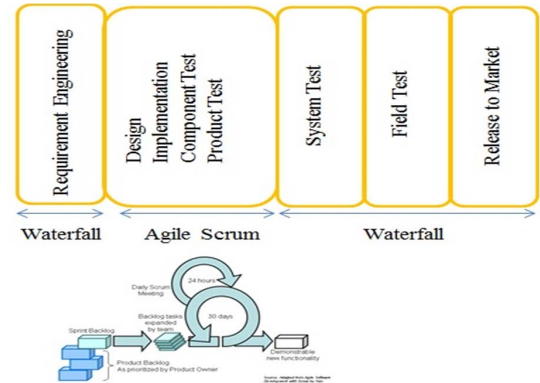


Figure 1 Hybrid Model

B. Backlog Grooming

Backlog Grooming follows layering and filtering process of waterfall model. We use multidimensional analysis approach which includes evaluating the business requirements against legacy systems of existing product family, existing reference competitor's system, technological advancement and Market needs. Once the requirement is validated through all these four validation points, it is moved to the Program Backlog. The Program Backlog now goes through the segregation process where every requirement is being evaluated on measures like whether the need is specific to the application or generic need for platform. Filtering process starts on segregated Program Backlog requirements. In the first level of filtering platform specific requirements, which are generic in nature, are selected and moved to the next level by filtering out application requirements. In the second level of filtering, only requirements which are synchronized with business goals are moved to the Release Backlog. All Release Backlog requirements have an absolute rank. Based on the program team capacity and project end date the final Release Backlog is created. All finalized high ranked Release Backlog requirements are also evaluated for non-functional requirements (NFR) with the help of scenario based approach. Availability of absolute ranked Release Backlog with identified NFR passes the readiness check for Agile Scrum execution.

C. Empower Scrum Team

1. Global Central Leadership

New set of roles were defined and introduced for program monitoring and control. Representatives from different engineering streams were chosen for driving the overall program. This team has representations from Architecture, Project Management, Product Portfolio Management, Product Testing, System Testing, Release Management, End User Training, and Product Documentation. Global Central Leadership team work hard to facilitate cross site impediments resolutions and expedite overall decision making process .

2. Scrum Team Organization

Scaling of Scrum has a huge impact on autonomy of the team and empowered team is the soul of Agile Scrum. Teams are organized with singular objective of “Make them independent and reduce dependency”. Scrum teams were assigned modules based on competency, domain know-how, and market proximity for support. Scrum teams have complete ownership for the assigned modules. Product Owners (14+) were assigned for the modules, so that Product Owners will help teams in meeting business objectives of the module. Program level Product Owner (Discipline Product Manager) ensures that product owners are working together to meet program level business objective. Global technical leads (14+) were empowered for taking technical decisions. Scrum Master (4+) has to facilitate inter team location issues resolution as well as also facilitate inter site team issue resolution by coordination and cooperation with other location Scrum Master. Central Leadership also plays a pivotal facilitation role in inter site issue resolution. Roles like Site Manager were introduced for resolving people management, infrastructure, procurement issues and logistical issues.

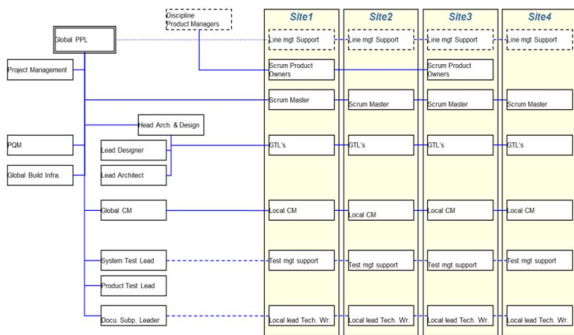


Figure 2 Program Organization

D. Model for Inter and Intra Team Communication

Communication is key to agile practices. Our communication model is designed on two levels, Horizontal and aggregated Vertical communication.

Horizontal communication is a communication which takes place within the function such as:

- Architects have weekly meeting with all the global technical leads (GTLs) to review design decision taken by the scrum teams. So all the stakeholders from different locations responsible for making technical decisions know the implications on their module as well as can suggest good design practices in order to make overall product design and architecture sustainable.
- Test Management (Product and System) will have their weekly meeting with their counterpart at different sites, over a Virtual Meeting, for deliberating on testing related progress, dependency and issue resolution.
- Build and configuration management team will have weekly meeting with their counterpart at different sites, over a Virtual Meeting, for deliberating topics on build management dependency and issue resolution.

- Documentation team will have weekly meeting with their counterpart at different sites, over a Virtual Meeting, for deliberating topics on documentation for dependency and issue resolution.
- Scrum Master will have communication with his team for plan progress and impediments.

Vertical Aggregated communication where all:

- Scrum Masters along with Global leads for documentation, Test Management (Product/System), Release Management, Architect, Global Project Lead, End User Training meet on weekly basis to discuss all location sprint delivery progress, dependencies and issue resolution.
- Global Leadership team also meets on weekly basis to review overall program progress and key impediments.
- Finally Central Global Leadership communicates overall progress to top management.

E. Sprint Plan – Global bottom-up planning

All Scrum Masters are responsible for their sprint plan creation for their respective teams based on the available capacity. Team Capacity is calculated based on the team's velocity. Every Scrum master shares their teams' capacity and Global project lead roll up the numbers of all the teams to decide the outline for the market release.

Sprint Plans have Theme/Epic and Stories. Every requirement is linked with Theme and for one requirement we will have only one Theme. The Theme can have multiple Epics and Epics can have multiple stories. All Theme/Epic and Stories are interlinked and inter team dependencies marked which enable teams to proactively adapt to the plan. Cross site dependencies are communicated in advance with correct linking with Theme/Epic/Story. Global Project Lead is responsible for rolling up the site specific sprint plan at program level and monitors the overall program progress and facilitates highlighting risk, if any team, is on a critical path.

F. Quality First

Since we are developing a platform on which future applications will be built and also can be used as a global product. Hence, it's imperative to have the mindset as well as measures for “Quality First”.

- Unique Infrastructure for Unit Testing by having two suits of Unit Tests V1 and V2 suits, where V1 test suits will not require platform support and V2 test suits will require platform support for execution. Both the test suits are completely automated and require very minimal manual intervention.
- Infrastructure for Static Code Analysis of all modules of platform
- Infrastructure for Build process, local as well as global
- Weekly Component test for delivered story in a week
- Monthly Product Tests for delivered Epic in a Sprint
- Monthly System Test once Theme is ready
- Weekly Automated and Manual Build Qualification Test (BQT) for ensuring consistency of delivery by all scrum team in a sprint

- Strong and Robust Automation infrastructure for various variability testing like testing product on different Operating Systems (OS), different Office Versions, Different combination of Anti-viruses and firewalls testing, Compatibility testing of various OS patches and different product version computability testing

G. Common Goal

When it comes to goal setting it binds all as we have an approach of having common goals on parameters like Project performance, quality targets, Processes and Budget Adherence. All functions share the same goal like Project management team which includes Scrum Masters and Global Project Lead share the same goals, Architect and GTLs, similar is with Test, release and documentation management.

VI. OPERATIONALIZED SCALED SCRUM

Operationalization holds the key to success and it also plays an important role in identifying what is working at ground level and what requires tweaking so that we get better results.

A. Hybrid Model – Process Tweaked for Excellence

During our journey we identified that we have to tweak our hybrid model for generating better utilization of available capacity and this in turn will lead to great results. Keeping this objective in mind we identified opportunity in modification of some of the hybrid model processes given below:

1) *Agile Approach for Requirement Engineering* –Hybrid model defines requirement engineering phase as waterfall but we executed the complete process with Agile philosophy of Inspect and Adapt with lean processes resulting in reduction of wait time. System testers were included in requirement discussion for documenting acceptance criteria on behalf of product owners, members from development team supported product owner in documenting the requirement which significantly impacted on over all requirement engineering process quality and schedule.

2) *Quality over Quantity* – We introduced the concept of “stop the line” which emphasizes on quality over quantity. If any Scrum team is having defects count above the ceiling level in a particular Sprint, we encourage the team to deliver less features than projected velocity and focus on quality. This includes closing existing defects and working on preventive mechanism to reduce defects for future sprints.

3) *Early System and Product Acceptance Test* – System test is supposed to start once all development is over and follow waterfall model but we made change to this process and included system test team inside Scrum by starting system test of requirements as and when they pass the product test and are ready for system test. We kept the dedicated system test phase unchanged. We also started early product acceptance test of the requirements which has passed system test. By using this approach, we get defect of system and product acceptance test early, resulting in positive impact on overall product quality.

B. Sprint Review and Reporting

Sprint review and reporting for us is putting robust and proactive inspect and adapt mechanism in place where we review sprint planning, performance of quality assurance mechanism and process adherence so that we are collectively reaching to target with predictability. The Measures put in place are given below:

1) *Weekly Review*: Weekly review meeting is chaired by global project lead where participants from global leadership team participate along with Scrum Masters to deliberate on following aspect of program

a) *Sprint Progress Review*: All Scrum Masters are responsible for reporting overall Sprint progress status (Scrum of Scrum) of their Scrum teams to the global project lead on the following parameters

- Highlight Impediments
- Status on what is getting delivered in this week
- Status on overall Sprint plan – On target or Not going to meet sprint target

- Weekly defect status (Open/Implemented)

b) *Progress on overall program quality*: Global Project lead presents the overall progress collective report of program which has the data from all Scrum teams. The Report has the following details:

- Weekly UT execution report
- Weekly Static Code Analysis report
- Program defect carpet along with its trend

c) *Product Test progress*: Global Product test lead provide status on the overall program product testing status, impediments if any.

d) *System Test progress*: Global System test lead provide status on the overall program system testing status, impediments if any.

e) *Product Documentation progress*: Product documentation lead provide status on the overall program product documentation status, impediments if any.

The objective of this meeting is to understand inter Scrum team dependencies, external impediments and how we can improve across site collaboration so that collectively we are achieving our program goal.

2) *Monthly Reporting*: Global Project lead is responsible to update overall program on monthly basis to the senior management on Schedule, Budget and Quality based on the input gathered over four weeks by the Scrum Masters

3) *Milestone Meeting* : Global project lead along with global leadership team present milestone readiness status on Scope, Schedule, Cost and Quality to the senior management who in-turn takes decision on granting proposed milestone.

C. Visualization for Transparency

1) *Tools Infrastructure for Scaling Scrum* : State of art tools infrastructure for planning and tracking, Test Management, Defect Management, Requirement Management are used. All these tools are interlinked and have bidirectional linking, like requirement is linked with Themes, Themes are linked with Test cases and if there is any defect, then the

defect is linked with the Theme. This linking is established across Theme/Epic and Stories. Components test defects are linked to stories, Product test defects are linked to Epics and System test defects are linked to Themes. Dependencies across Scrum teams are also managed in the same tool which helps Scrum Master to proactively adapt their Sprint plan.

2) Set Configured as Query for Quick Gap Identification :

Queries for general guidelines for tool usage are created to identify if there is any gap. Some of the gaps can be like story is planned in a sprint and if the status is in “In-progress” even after the sprint, then it will come out as gap, absent of requirement to theme linking will be marked as gap, no child of theme or orphan Story/Theme/Epic will be marked as gap etc

3) Dashboard for Program and Scrum Team

Dashboards are available from individual Scrum teams which can be rolled up to the programs level sprint on sprint which ensures transparency across all roles. Below mentioned picture is a snapshot of program level dashboard for sprint progress.

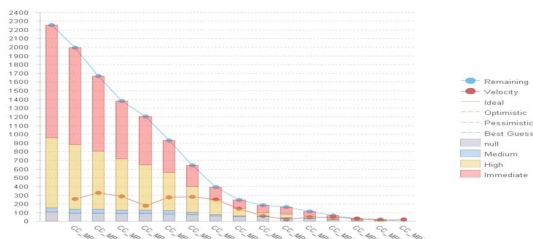


Figure 3: Program Sprint Progress Dashboard

D. Proactive Risk Management

Focus for risk management is a systematic approach for minimizing exposure to potential losses for individual teams and overall program. The primary sources for risk are lessons learnt from previous version, Sprint retrospective findings. Global Project Lead facilitates continuous assessment of what could go wrong and ensures high-priority risks are addressed. Top five risks are presented to top management every month and tracked very closely. Technical risks are managed and tracked by Architects as technical debts.

E. Product Release Management

Continuous integration is one of the important Scrum practices. We have the following build process:

1) *Daily Builds* : Happens over night everyday for each team.

2) *Weekly Builds*: Three kinds of weekly builds are created every week. Team build is created when scrum team finalizes its delivery. Site build is invoked when all scrum teams builds are created successfully and when all site builds are created, global build is invoked which is consumed by the components, product and system test teams.

F. *Continuous Improvement* : We believe in continuous improvement, hence we keep refining our processes based on our failures while execution.

VII. CHALLENGES AND LESSONS LEARNED

1) Sprint coordination:

a) Make sure dependencies are identified on time and the sprint planning for all teams is done accordingly.

b) Make sure we are working on the highest ranked requirements, not starting too many at the same time, but closing first the existing ones before starting new ones (and this assuring the whole team workload). Target is to focus on the important features and to be able to verify as soon as possible this functionality and reduce the risk with it

2) Avoid late change requests by requesting early acceptance tests by PO for the closed features

3) Coordination between development teams, documentation and training, assures the know how transfer and the update of the last changes to all the stakeholders

4) Find a balance between feature implementation and quality

5) Availability of groomed Backlog is a key to success otherwise teams have to wait for work and it will have an impact on cost and capacity utilization

6) Managing tool ecosystem along with velocity is overwhelming and requires drive and motivation from top

7) Management should be open to accepting failures because scaling is like juggling act and failures are bound to happen

8) Scaling requires robust tools ecosystem in place for transparency and visualization

9) Scaling has high entry investment cost and initially may not seem to be a correct choice

10) Failure should be looked at as a short term hindrance for long term benefits

11) Requires committed management and motivated leadership for successful implementation

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